DAY 21 | HANDS-ON ACTIVITIES

# GenAI Intro and Prompt Engineering: Problem Statements

Seven problems. Every notebook cell runs offline; add an OPENAI_API_KEY to see real model behaviour.

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**How to work:** read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

## Problem 1: Count the Tokens

**Level:** Warm-up | **Concepts:** tokens, context window

Tokenise three texts of different lengths (a tweet, an email, a page of a policy) and estimate how many pages would fit into a 128,000 token context window.

### Sample run / expected result

```
Tweet: 31 tokens
Email: 212 tokens
Policy page: 640 tokens
Approx. pages in 128k: 200
```

### Hints

- Use tiktoken if available, otherwise about 4 characters per token.

## Problem 2: Temperature Explorer

**Level:** Easy | **Concepts:** softmax, sampling

Given logits for 5 candidate next words, print the probabilities at temperature 0.2, 1.0 and 2.0, and sample 10 words at each temperature.

### Sample run / expected result

```
T=0.2: 'sunny' 0.97
T=2.0: 'sunny' 0.38, 'rainy' 0.24 ...
```

### Hints

- probs = exp(logits / T) / sum(exp(logits / T))

## Problem 3: Attention by Hand

**Level:** Medium | **Concepts:** Q, K, V, softmax

Using small made-up vectors for the words in "bank of the river", compute the attention weights and show which word "bank" attends to most.

### Sample run / expected result

```
bank attends most to: river (0.41)
```

### Hints

- Make the vectors for bank and river similar on purpose.

## Problem 4: Prompt Makeover

**Level:** Easy | **Concepts:** prompt anatomy

Take the prompt "write about our product" and rewrite it with role, task, context, format, examples and constraints. Compare the outputs (online) or the prompts (offline).

### Sample run / expected result

```
Before: 5 words
After: role + task + 3 facts + format (3 bullets, under 60 words) + 1 example
```

### Hints

- Specific beats long.

## Problem 5: Few-shot Classifier

**Level:** Medium | **Concepts:** few-shot prompting

Build a sentiment classifier prompt with three examples, run it on 8 customer messages, and measure accuracy against your own labels.

### Sample run / expected result

```
Accuracy: 7/8
```

### Hints

- Ask for exactly one word so you can compare easily.

## Problem 6: ReAct with Tools

**Level:** Challenging | **Concepts:** ReAct loop, tool calling

Extend the notebook's ReAct agent with a currency converter tool and answer: "What is the GST on P-200 in Japanese yen?"

### Sample run / expected result

```
Thought: need price -> lookup_price
Thought: need GST -> calculator
Thought: convert -> convert_currency
Final Answer: ...
```

### Hints

- Limit the loop to 6 steps to avoid runaway agents.

## Problem 7: Tree of Thought Planner

**Level:** Stretch | **Concepts:** explore, evaluate, choose

Generate three different plans for reducing customer support response time, score each on impact, effort and risk, and choose one with a short justification.

### Sample run / expected result

```
Plan B (FAQ bot + routing) scores 8.5/10: high impact, medium effort, low risk
```

### Hints

- Score with clear criteria; ask the model to justify each score.

*Ganbatte (頑張って): do your best*